

Amdahl's Law

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1 Introduction

This document discusses Amdahl's Law, a fundamental principle in parallel computing that helps predict the theoretical speedup in latency when using multiple processors.

1.1 Performance Metrics

Considering $T(p)$ the time to solve a problem using p processors, we define:

- **Speedup:** $s(p) = \frac{T(1)}{T(p)}$, $0 < s(p) \leq p$
- **Efficiency:** $e(p) = \frac{s(p)}{p}$, $0 < e(p) \leq 1$

2 The Law

Amdahl's Law can be expressed as a classical formula:

$$s(p) = \frac{1}{S + \frac{(1-S)}{p}}$$

where:

- $s(p)$: is the theoretical speedup
- S : is the proportion of sequential code (non-parallelizable)
- $1 - S$: is the proportion of parallelizable code
- p : is the number of processors

2.1 Ideal case ($S = 0$)

Implies that all code is parallelizable. The speedup becomes:

$$s(p) = \frac{1}{\frac{1}{p}} = p$$

Maximum speedup equals the number of processors. Each processor contributes fully to the speedup.

2.2 Theoretical limit case ($p \rightarrow \infty$)

By increasing the number of processors to infinity, the speedup reaches its theoretical limit:

$$\lim_{p \rightarrow \infty} s(p) = \frac{1}{S}$$

Even though the number of processors increases indefinitely, the speedup is limited by the sequential portion of the code, encountering a bottleneck.

2.3 Real case ($S = \alpha p$)

Where α is a constant $\alpha \leq 0.1$ representing the overhead per processor. The speedup becomes:

$$s(p) = \frac{1}{\alpha p + \frac{1-\alpha p}{p}} = \frac{p}{1 - \alpha p + \alpha p^2}$$

There is no free lunch: as we add more processors, the overhead increases linearly, impacting the overall speedup.

2.3.1 For α small

When α is small we do not have a significant overhead.

$$s(p) \approx p$$

2.3.2 For α large

When α is large, the overhead dominates. The more we add processors, the less benefit we get!

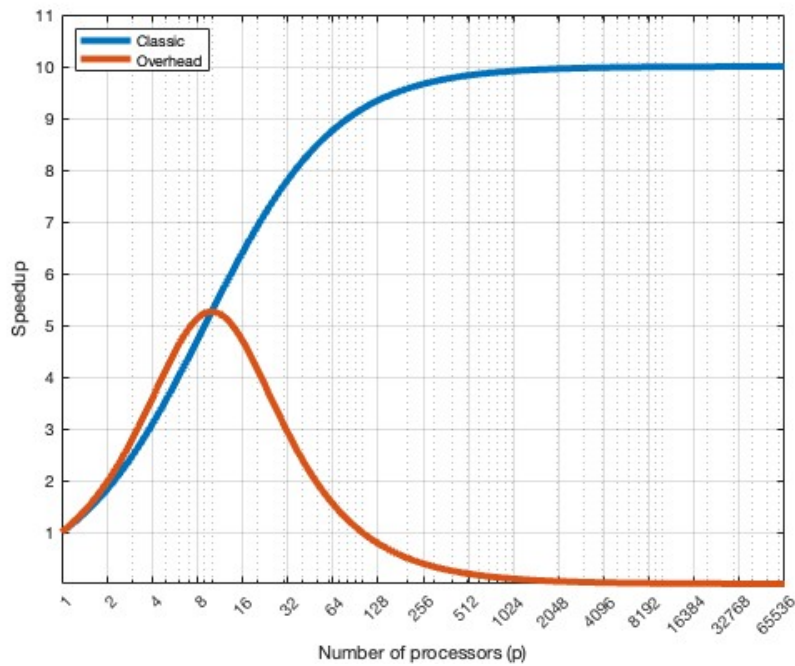
$$s(p) \approx \frac{p}{\alpha p^2} \approx \frac{1}{\alpha p}$$

2.4 Examples

The following examples have been obtained using Matlab scripts available in the repository.

Classical vs Overhead

- The proportion of the parallelizable is 95% of code.
($S = 0.1$, $P = 1 - S = 0.9$)
- Overhead per processor
 $\alpha = 0.01$



Parallel speedup comparison

It is useful for understanding the limits of parallelization according to Amdahl's law: it highlights that simply increasing the number of processors is not enough if the sequential portion of the code is significant

